

INCIDENT REPORT

Author: Israel Olawuni

Date: 25 June, 2026.

1. Incident Summary

What was the problem?

Windows Server (DC01) and Ubuntu Server (dns01) could not communicate with each other over the 10.10.10.0/24 lab network. Ping in both directions failed with 100% packet loss.

When was it detected?

Detected during Task 2.2.1 when the network_info.py script was run on DC01 and showed 192.168.100.10 instead of the expected 10.10.10.10.

When was it resolved?

Fully resolved on 07 July 2026 when bidirectional ping between 10.10.10.10 and 10.10.10.20 returned 0% packet loss.

What was the impact?

No communication possible between DC01 and dns01 on the target network

2. Diagnosis

Root Cause Identified

The failure had two compounding causes operating at different layers:

1. Ubuntu Server's Adapter 2 was attached to NAT instead of Host-only Adapter in VirtualBox settings.
2. IP configuration: On DC01, the static IP 10.10.10.10 was repeatedly applied to Ethernet instead of Ethernet 2.

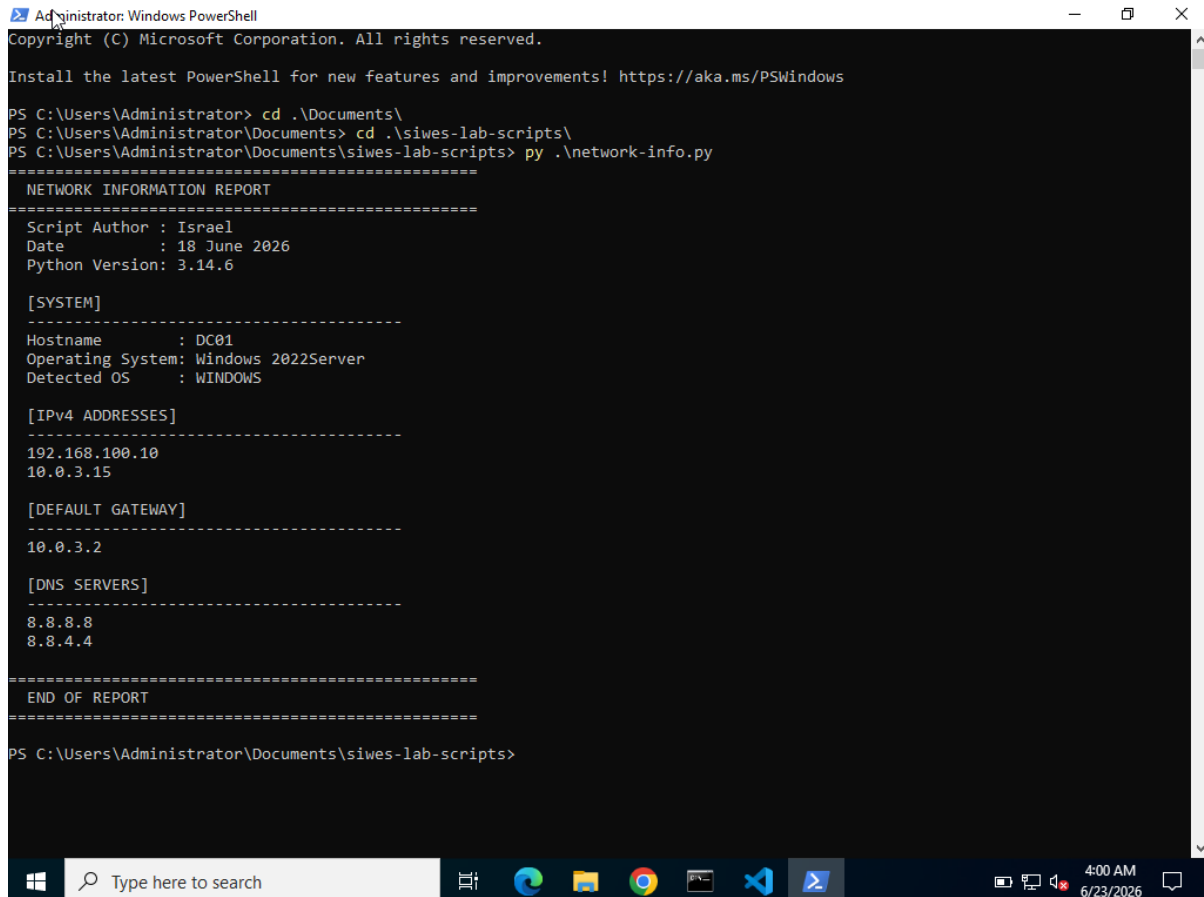
How It Was Discovered

Step 1: Running network_info.py on DC01 showed:

[IPv4 ADDRESSES]

192.168.100.10

10.0.3.15



```
Administrator: Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Administrator> cd .\Documents\
PS C:\Users\Administrator\Documents> cd .\siwes-lab-scripts\
PS C:\Users\Administrator\Documents\siwes-lab-scripts> py .\network-info.py
=====
NETWORK INFORMATION REPORT
=====
Script Author : Israel
Date          : 18 June 2026
Python Version: 3.14.6

[SYSTEM]
-----
Hostname      : DC01
Operating System: Windows 2022Server
Detected OS   : WINDOWS

[IPv4 ADDRESSES]
-----
192.168.100.10
10.0.3.15

[DEFAULT GATEWAY]
-----
10.0.3.2

[DNS SERVERS]
-----
8.8.8.8
8.8.4.4

=====
END OF REPORT
=====
PS C:\Users\Administrator\Documents\siwes-lab-scripts>
```

Neither address matched the expected 10.10.10.10.

Step 2: Running network_info.py on dns01 showed:

[IPv4 ADDRESSES]

10.0.2.15

192.168.100.20

```

slwesadmin@dns01:~$ cd Document
slwesadmin@dns01:~/Document$ chmod +x network_diagnostic.sh
slwesadmin@dns01:~/Document$ ./network_diagnostic.sh
=====
NETWORK INFORMATION REPORT
=====
Script Author : Israel
Date          : 24 June 2026
Shell         : Bash 5.3.9(1)-release

[SYSTEM]
-----
Hostname      : dns01
Operating System: Linux 7.0.0-22-generic

[NETWORK ADAPTER STATUS]
-----
Adapters UP:
- enp0s3
- enp0s8

[IPv4 ADDRESSES]
-----
10.0.2.15
192.168.100.20

[DEFAULT GATEWAY]
-----
10.0.2.2

[DNS SERVERS]
-----
192.168.144.94
8.8.8.8

[DIAGNOSTIC SUMMARY]
-----
Status: ALL CHECKS PASSED
The network configuration appears healthy.

=====
END OF REPORT
=====
slwesadmin@dns01:~/Document$

```

Neither address matched 10.10.10.20.

Step 3: Ping test (Task 2.2.3):

Windows → Ubuntu: Destination Host Unreachable (100% loss)

Ubuntu → Windows: Network is unreachable (100% loss)

Step 4: After initial fix attempts failed, VirtualBox settings were opened and MAC addresses were cross-referenced against Get-NetAdapter (Windows) and ip addr show (Ubuntu) output. This revealed:

VM	VirtualBox Adapter	MAC	OS Interface	Configured Role	Actual Role
DC01	Adapter 1	08-00-27-9A-FD-B7	Ethernet 2	Host-Only	Host-Only
DC01	Adapter 2	08-00-27-C7-CE-73	Ethernet	NAT	NAT

dns01	Adapter 1	08-00-27-A9-40-64	enp0s3	NAT	NAT
dns01	Adapter 2	08-00-27-1B-2C-4A	enp0s8	Should be Host-Only	NAT

The VirtualBox misconfiguration on dns01 Adapter 2 was identified as the deepest layer of the failure.

Evidence Collected

Evidence Item	Source	Finding
Windows script output	Task 2.2.1	No 10.10.10.x address on any adapter
Ubuntu script output	Task 2.2.2	No 10.10.10.x address on any adapter
Ping results (both directions)	Task 2.2.3	100% loss, "unreachable" error type
Get-NetAdapter MAC table	Task 2.2.5	Adapter name $\neq$ adapter slot order
VirtualBox Network settings (DC01)	Task 2.2.5	Adapter 1 = Host-Only, Adapter 2 = NAT
VirtualBox Network settings (dns01)	Task 2.2.5	Adapter 1 = NAT, Adapter 2 = NAT (wrong)
ip addr show after Ubuntu OS fix	Task 2.2.5	10.10.10.20 on enp0s8, but ARP still failing

3. Resolution

Step-by-Step Fix Applied

Diagnosis from Task 2.2.4 was confirmed still valid. The predicted fix (assign correct static IPs to both servers) remained correct.

1. VirtualBox: Reconfigure dns01 Adapter 2

Setting	Before	After
DNS01 Adapter 2 → Attached to	NAT	Host-only Adapter
DNS01 Adapter 2 → Name		VirtualBox Host-Only Ethernet Adapter

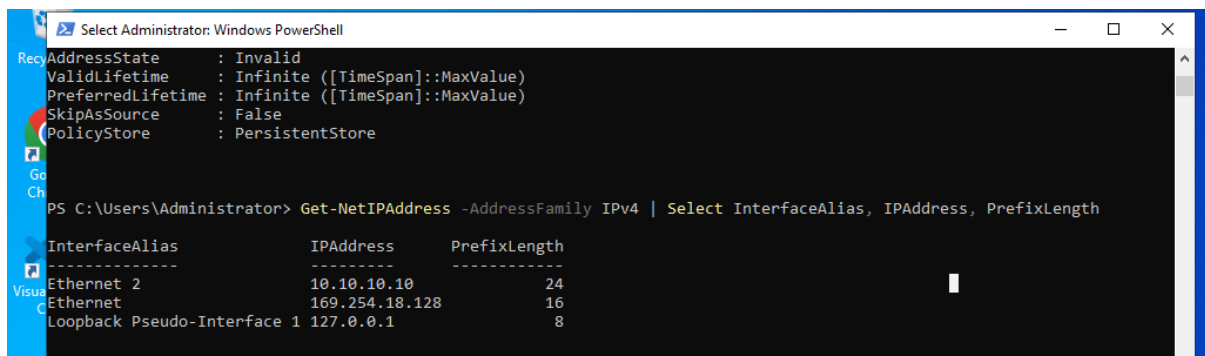
2. Windows: Move Static IP to Correct Adapter

Commands executed:

Remove-NetIPAddress -InterfaceAlias "Ethernet" -Confirm:\$false

New-NetIPAddress -InterfaceAlias "Ethernet 2" -IPAddress 10.10.10.10 -PrefixLength 24

Get-NetIPAddress -AddressFamily IPv4 | Select InterfaceAlias, IPAddress, PrefixLength



```
Select Administrator: Windows PowerShell
RecvAddressState : Invalid
ValidLifetime : Infinite ([TimeSpan]::MaxValue)
PreferredLifetime : Infinite ([TimeSpan]::MaxValue)
SkipAsSource : False
PolicyStore : PersistentStore

PS C:\Users\Administrator> Get-NetIPAddress -AddressFamily IPv4 | Select InterfaceAlias, IPAddress, PrefixLength

InterfaceAlias      IPAddress      PrefixLength
-----
Ethernet 2          10.10.10.10    24
Ethernet            169.254.18.128 16
Loopback Pseudo-Interface 1 127.0.0.1      8
```

3. Ubuntu: No Additional Changes Required

Ubuntu's Netplan configuration already had 10.10.10.20/24 on enp0s8. After VirtualBox Adapter 2 was changed to Host-Only, enp0s8 automatically became reachable from DC01.



```
slwesadmin@dns01:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:a9:40:64 brd ff:ff:ff:ff:ff:ff
    altname enx00027a94064
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 86359sec preferred_lft 86359sec
    inet6 fd17:625c:f037:2:a00:27ff:fea9:4064/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86361sec preferred_lft 14361sec
    inet6 fe80::a00:27ff:fea9:4064/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:1b:2c:4a brd ff:ff:ff:ff:ff:ff
    altname enx000271b2c4a
    inet 10.10.10.20/24 brd 10.10.10.255 scope global enp0s8
        valid_lft forever preferred_lft forever
    inet6 fe00::a00:27ff:fea9:2c4a/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
```

Challenges Encountered

1. Windows adapter naming (Ethernet vs Ethernet 2) does not match VirtualBox slot order
2. Ubuntu Adapter 2 configuration

4. Verification

Script Output

Ubuntu

```
siwesadmin@dns01:~$ cd Document
siwesadmin@dns01:~/Document$ python3 network_info.py
=====
NETWORK INFORMATION REPORT
=====
Script Author : Israel
Date          : 18 June 2026
Python Version: 3.14.4

[SYSTEM]
-----
Hostname       : dns01
Operating System: Linux 7.0.0-22-generic
Detected OS    : LINUX

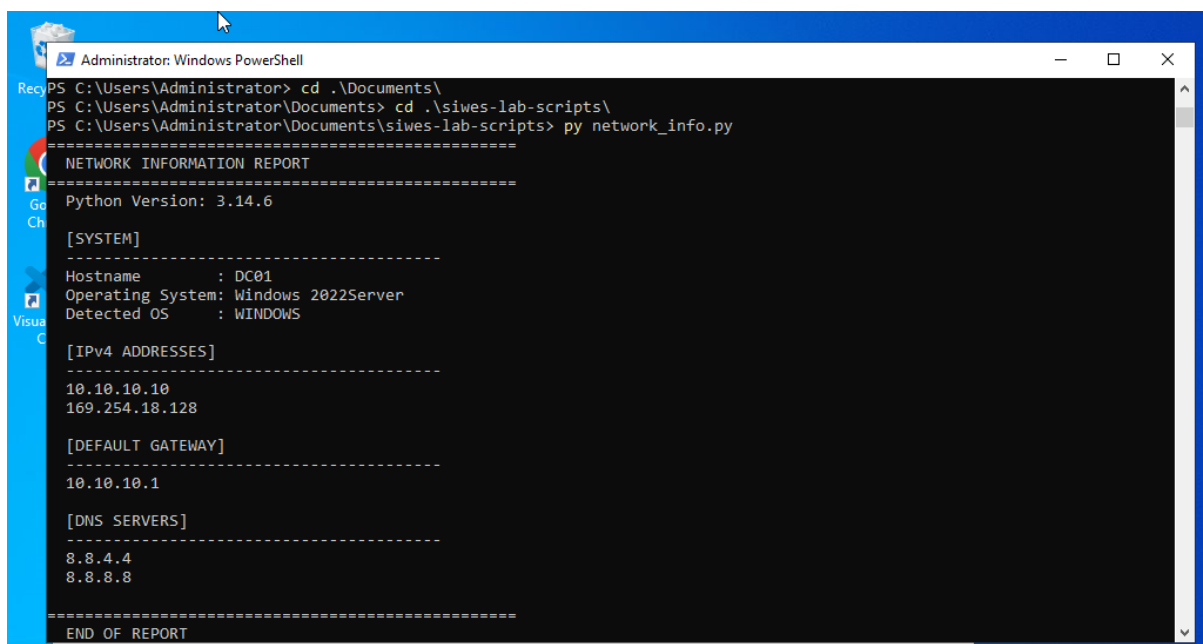
[IPv4 ADDRESSES]
-----
10.0.2.15
10.10.10.20

[DEFAULT GATEWAY]
-----
10.10.10.1

[DNS SERVERS]
-----
10.0.2.3
8.8.8.8

=====
END OF REPORT
=====
siwesadmin@dns01:~/Document$
```

Windows

A screenshot of a Windows PowerShell window titled "Administrator: Windows PowerShell". The window shows the execution of a Python script named "network_info.py". The script outputs a "NETWORK INFORMATION REPORT" with details about the system, IP addresses, default gateway, and DNS servers. The report is formatted with dashed lines separating sections. The PowerShell prompt shows the user navigating to the directory "C:\Users\Administrator\Documents\siwes-lab-scripts\" and running the command "py network_info.py".

```
Administrator: Windows PowerShell
PS C:\Users\Administrator\> cd .\Documents\
PS C:\Users\Administrator\Documents> cd .\siwes-lab-scripts\
PS C:\Users\Administrator\Documents\siwes-lab-scripts> py network_info.py
=====
NETWORK INFORMATION REPORT
=====
Python Version: 3.14.6

[SYSTEM]
-----
Hostname       : DC01
Operating System: Windows 2022Server
Detected OS    : WINDOWS

[IPv4 ADDRESSES]
-----
10.10.10.10
169.254.18.128

[DEFAULT GATEWAY]
-----
10.10.10.1

[DNS SERVERS]
-----
8.8.4.4
8.8.8.8

=====
END OF REPORT
=====
```

Ping Tests

Windows to Ubuntu:

```
Pinging 10.10.10.20 with 32 bytes of data:
Reply from 10.10.10.20: bytes=32 time=8ms TTL=64
Reply from 10.10.10.20: bytes=32 time=2ms TTL=64
Reply from 10.10.10.20: bytes=32 time=2ms TTL=64
Reply from 10.10.10.20: bytes=32 time=5ms TTL=64

Ping statistics for 10.10.10.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 8ms, Average = 4ms
PS C:\Users\Administrator>
```

Ubuntu to Windows:

```
sluesadmin@dns01:~$ ping -c 4 10.10.10.10
PING 10.10.10.10 (10.10.10.10) 56(84) bytes of data.
64 bytes from 10.10.10.10: icmp_seq=1 ttl=128 time=3.46 ms
64 bytes from 10.10.10.10: icmp_seq=2 ttl=128 time=2.25 ms
64 bytes from 10.10.10.10: icmp_seq=3 ttl=128 time=2.11 ms
64 bytes from 10.10.10.10: icmp_seq=4 ttl=128 time=3.13 ms

--- 10.10.10.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3008ms
rtt min/avg/max/mdev = 2.107/2.736/3.460/0.571 ms
sluesadmin@dns01:~$ _
```

5. Before vs After Comparison

Test	Command	Expected	Result	Pass/Fail
Windows IP correct	py network_info.py	Shows 10.10.10.10	10.10.10.10	Pass
Windows Gateway correct	py network_info.py	Shows 10.10.10.1	10.10.10.1	Pass
Ubuntu IP correct	python3 network_info.py	Shows 10.10.10.20	10.10.10.20	Pass
Ubuntu Gateway correct	python3 network_info.py	Shows 10.10.10.1	10.10.10.1	Pass
Windows → Ubuntu ping	ping 10.10.10.20	0% loss, replies received	4/4 received, avg 4ms	Pass

Ubuntu → Windows ping	ping -c 4 10.10.10.10	0% loss, replies received	4/4 received, avg 2.7ms	Pass
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6. Lessons Learned

What would you do differently next time?

1. Verify the hypervisor layer first, before touching OS configuration.
2. Create a reference table mapping VirtualBox slot to MAC. What monitoring could have detected this earlier?

What to Monitor	How	Would Have Caught
ARP table entries	arp -a (Windows) / ip neigh (Ubuntu) after IP assignment	An empty ARP table after assigning an IP immediately signals the adapter is not on the right network
Adapter state in VirtualBox	Check Settings before booting VMs	DNS01 Adapter 2 being NAT would have been visible before any commands were run
Expected-value comparison in diagnostic script	Add --expected-ip flag that fails if actual IP doesn't match	Would have caught the APIPA and wrong-adapter states immediately instead of reporting "PASSED"
ip route / Get- NetRoute output	Include in diagnostic script	Would have shown missing route to 10.10.10.0/24 explicitly

What documentation should be updated?

1. Lab setup guide that includes the MAC cross-reference method.
2. network_diagnostic.sh README
3. network_info.py README

What preventive measures can be implemented?

1. Script improvement

2. Document the permanent MAC-to-role mapping for each VM so adapter identity never needs to be re-investigated:

VM	VirtualBox Adapter	MAC	OS Interface	Role
DC01	Adapter 1	08-00-27-9A-FD-B7	Ethernet 2	Host-Only
DC01	Adapter 2	08-00-27-C7-CE-73	Ethernet	NAT
dns01	Adapter 1	08-00-27-A9-40-64	enp0s3	NAT
dns01	Adapter 2	08-00-27-1B-2C-4A	enp0s8	Host-Only